

QK2 Decision Register

The dossier's internal consistency matrix, reconciled to **QuietKey Core Architecture v1**. Each decision is **settled (product direction), deferred, rejected, or open (stop-ship)**. This register is subordinate to QuietKey Core Architecture v1, to QK2-04, and to the QK2-HW-P0/RV3 reviews; where it disagrees with v1, v1 is correct.

PROJECT STATUS – SELECTED PRODUCT ARCHITECTURE; IMPLEMENTATION VALIDATION REMAINS STOP-SHIP

QK2 is an interchangeable, air-gapped Bitcoin signer disguised as a retro calculator. It stores no wallet secret at rest. The wallet is native **P2WSH 2-of-3** with independent Seeds A/B/C. **A1** is an ordinary printed webpage whose source-URL-looking footer contains authenticated ciphertext for 256-bit Seed A. **A2** is freshly generated during provisioning, copied to both bearer cards B and C, and wiped from the terminal. **B** carries signer B as a provisioned account xprv ($m/48'/0'/0'/2'$) plus A2 and descriptor D; **C** carries independent signer C (account xprv) plus the same A2 and D, and belongs to a configurable recovery/inheritance kit. The applet holds no BIP39 mnemonic — the device derives B/C from independent 256-bit seeds and provisions only the least-authority account xprv. There is **no card PIN or terminal-specific pairing**. Normal signing is A1+B; recovery is A1+C or B+C. Seeds are fixed at **24-word / 256-bit**. **QR (BBQr) and microSD (binary PSBT v0)** transport are both required on the existing mechanical design. Descriptor confidentiality at rest, quantum-exposure minimization, and future PQ migration are design requirements. Card feasibility, A1 encoding/capture, QR performance, signer implementation, recovery rehearsal, and QK2-HW-P0/RV3 evidence remain blockers. Reference vectors are not implementation validation.

01 CANONICAL PRODUCT

Four logical elements, three custody objects, one empty device

CloakVault is defined by four inseparable ideas — **stealth · minimalism · strong encryption · hiding in plain sight**. Stealth is a first layer of defence, never a cryptographic factor and never a substitute for the 2-of-3 threshold. Every visible artifact must look ordinary.

01-3 threshold. Every visible artifact must look ordinary.

ELEMENT	PHYSICAL FORM	CONTENTS	FUNCTION
A1	ordinary printed webpage	authenticated ciphertext of 256-bit Seed-A entropy, carried in a footer resembling the page's source URL	restores Seed A transiently; safe to leave visibly among ordinary papers because it is encrypted
A2	not a daily object; duplicated onto both cards	fresh 256-bit random key made at provisioning	decrypts A1; it is not a Bitcoin signing key
B	bus-pass-like smart card	provisioned account xprv (signer B, $m/48'/0'/0'/2'$) + A2 + descriptor D; no BIP39 mnemonic on the card	normal card signer
C	second bus-pass card + recovery/inheritance kit	independent signer C as an account xprv + the same A2 + D; recovery packaging per kit tier	replaces A1/A or B in a one-factor-loss event
Device	retro-calculator terminal	no persistent wallet secret	interchangeable scanner, validator, display, signer host

The terminal must not contain A2 before setup. At provisioning it generates A2 with A1, writes the same A2 to B and C, verifies both copies, and wipes A2 and all seeds from terminal storage/RAM at ceremony completion.

The wallet is selected: `wsh(sortedmulti(2, A, B, C))`

normal use A1 + B → decrypt A with A2 → sign A + B
B lost/unavailable A1 + C → decrypt A with A2 → sign A + C
A1 lost B + C → sign B + C; A is bypassed entirely
C lost A1 + B → sign A + B, then sweep to a fresh kit
terminal lost replace the terminal; no key recovery is needed

Temporary unavailability of any ONE of A1, B, C, or the terminal is tolerated. Any LOST or suspected-compromised A1/B/C is spent once with a surviving pair, then swept into a completely fresh A'/B'/C'/A2'/D' kit – a factor is never "reissued" (replacing a key changes the descriptor and the funded wallet). Irrecoverable loss of TWO signing factors can still make the wallet unrecoverable – stated plainly.

B and C are deliberately **bearer objects**. There is no card PIN, OAC, document-derived activation secret, duress PIN, retry counter, or lockout. Physical custody of a card is its authorization; the Bitcoin 2-of-3 policy is the spending control. This matters most for C: the emergency/inheritance kit must work after a long interval with no forgotten PIN, vendor service, account, clock, or challenge-response.

02 THE MATRIX

Every decision, its state, and the documents that must reflect it

SETTLED product direction (not implementation evidence) DEFERRED out of v1 REJECTED appendix only OPEN stop-ship

DECISION	STATE	WHERE
Wallet policy = native P2WSH 2-of-3 wsh(sortedmulti(2,A,B,C)) ; m/48'/0'/0'/2'/{0,1}/* ; mainnet-only; independent seeds; A1+B normal, A1+C or B+C recovery; timelock/spare-B/single-sig and the retired 2-of-2 spine are rejected alternatives	SETTLED	00·01·03·04·05·08
Seed strength = 24 words / 256-bit fixed for A/B/C; A2 independently 256-bit; empty/absent BIP39 passphrase is fixed; 12-word/128-bit survives only as a labelled unit/standards vector, never a product mode	SETTLED	00·02
Camouflage is a system requirement calculator terminal, A1 as an ordinary webpage, bus-pass cards, neutral kits — affects artifact, format, UI, labels, packaging	SETTLED	all
A1 = cloaked, encrypted webpage encryption MANDATORY; Seed A as 32-byte entropy; source-URL-looking footer; no naked BIP-39, no conspicuous token/label	SETTLED	00·01·05·07
A2 lifecycle fresh 256-bit at provisioning; duplicated to B and C; wiped from terminal; exportable from a bearer card by design; not a Bitcoin key	SETTLED	00·01·05·06·08
No card PIN / OAC / duress / lockout B and C are bearer objects; physical custody = authorization; the 2 of 3 threshold is the control	SETTLED	00·01·06·08

B and C are bearer objects, physical custody = authorization, the 2-of-3 threshold is the control		
Device interchangeable no card↔terminal pairing; no terminal static key as a recovery dependency; terminal possession is not a spending factor	SETTLED	00·01·06·08
QR + microSD both required SD = binary PSBT v0 only (no base64/hex autodetect); QR = BBQR (file type P , uncompressed base32) — the one selected framing, with SeedQR/UR2/Specter removed as candidates; both required on the EXISTING mechanics; no mechanical redesign; BBQR performance on the fixed mechanics stays gated	SETTLED	00·01·03·04·08
Camera is mode-locked two uses only — A1 footer capture and PSBT QR; never auto-detect a seed QR or accept arbitrary QR classes	SETTLED	01·04·07
Complete descriptor D survives independently two checksummed receive/change descriptors (the conservative authority) with full origins/threshold/network/derivation/version/birthday/first-addresses; D lives on BOTH B and C plus one redundant A2-encrypted cloaked kit copy, so every valid pair (always containing a card) can recover the wallet map; <code>wallet_id = SHA256(recv_desc 0x00 change_desc)</code>	SETTLED	00·03·05·08
D is privacy-sensitive & quantum-sensitive not "public, copy widely"; offline+encrypted at rest; no cloud sync. Coordinator disclosure is TIERED: Simple Recovery & Inheritance MAY export the full watch-only descriptor to a supported coordinator; only Quantum Shelter is address-packs-only — there is no universal "addresses-only, never descriptor" rule	SETTLED	03·04·05·08
Quantum-prepared posture dormant P2WSH keys/descriptors offline; no address reuse; NOT Taproot for the vault; monitor BIP-360/P2MR; migrate when a reviewed PQ path deploys	SETTLED	01·03·04·05·08
PSBT v0 signing profile 2-of-3 corpus, three valid pairs; SIGHASH_ALL only; verify existing partial signatures and add only required fields (no gratuitous rewrite); checked arithmetic with explicit absolute fee + sat/vB, versioned warnings, and a conservative absolute emergency ceiling (no universal 10%-of-spend fatal rule); unknown/unrenderable = fatal; every completed local session returns a final raw tx (finalized PSBT where the tier/coordinator needs it); Quantum-Shelter returns raw tx only	SETTLED	01·04
Signing = deterministic RFC6979 ECDSA + strict DER + low-S terminal uses exact deterministic RFC6979 on pinned libsecp256k1; strict DER, low-S, verify before release; NOT "hedged RFC6979"; low-R is an optional size-only optimization; card signature construction is a Gate-B capability to specify and test	SETTLED	01·03·04·06·08
Kit tiers customize packaging, not crypto Simple Recovery · Inheritance · Quantum Shelter — same threshold/word-count/no-PIN/script/entropy; differ in packaging, instructions, redundancy, D placement, and coordinator disclosure (full watch-only descriptor for Simple/Inheritance vs address packs only for Quantum Shelter)	SETTLED	03·05·08
Kit construction = merge into a fresh kit	SETTLED	00·01·05·08

Loss semantics = sweep into a fresh kit any lost or suspected-compromised A1/B/C → spend once with a surviving pair, then provision a fresh A'/B'/C'/A2'/D' and sweep all funds; a factor is never "reissued" — replacing any key changes the descriptor and the funded wallet; lost terminal alone needs no on-chain rotation	SETTLED	00·01·05·00
Card provisioning = least-authority account xprv device derives Seed B/C from independent 256-bit entropy, then provisions only the account xprv at <code>m/48'/0'/0'/2'</code> (+ chain code + origin fingerprint) with A2 and D; the applet implements no BIP39/PBKDF2; mnemonic/root is wiped after account-xpub read-back and never lives on the card	SETTLED	02·06·08
A1 crypto capsule frozen (design selected) ChaCha20-Poly1305 (RFC 8439), NOT AES-GCM-SIV; doc key = <code>HKDF-SHA256(IKM=A2, salt=wallet_id, info="QuietKey/A1/v1")</code> ; fresh stored 96-bit nonce; 128-bit tag; AAD = <code>magic/coding+crypto versions/network/wallet_id</code> ; 32-byte Seed-A plaintext; no plaintext released before tag verification; an identical reprint reuses the same capsule	SETTLED	05·07
Implementation = Rust core + narrow libsecp256k1 FFI Rust for the core, parsers, state machines, recovery codec, and transport decoding; narrow pinned C FFI only for libsecp256k1; C is a measured fallback (Gate C), not a default; Python/JS forbidden at the secret boundary; three-process split qk-core (trusted) / qk-io (unprivileged broker) / qk-decoy (calculator)	SETTLED	01·04·06·08
PSBT v2 · anti-exfil / sign-to-contract · miniscript timelock · standard SCP11 card channel (later hardening only)	DEFERRED	01·03·04·06·08
Timelock · cloned-B · encrypted-B-export · single-sig · card PIN/OAC · terminal pairing · custom card secure channel (bespoke ECDH/AES/HMAC or fleet key) · plaintext-A backup · retired 2-of-2 spine	REJECTED	05·06·07
A1 print codec physical constants & scan-back crypto capsule is frozen (see the settled row above); still OPEN/Gate-A: shortened RS over GF(256) → OCR-safe base32 → checksum → source-URL path parameters, font, footer length, OCR erasure thresholds, and real print/photocopy/scan-back performance; RS correction is never authentication — only a verifying AEAD tag under the correct wallet_id accepts a candidate	OPEN	07
Card model & capability proof (Gate B) secp256k1 derive/ECDSA, account-xprv provisioning + role/wallet-id binding, A2 export, atomic tear/reset, timing, flash endurance, random source, power-cut on the actual target card; card signature construction to specify and test; the card bus is inside the trusted terminal boundary in v1 (no custom secure channel); ship a published APDU spec, an open CCID/ISO-7816 rescue CLI, and a reproducible rescue image	OPEN	06
QR capture/display/RAM performance on unchanged mechanics	OPEN	01·04
Entropy source characterization & end-to-end reachability proof terminal conditions all four values via QKEC-1 (HKDF-SHA256), cards contribute RNG samples only in Multi-source mode; build-time symbol/reachability checks + fault injection + a reproducible pure-physical mode of exactly 100 d6 rolls (the ColdCard lesson); terminal health checks stay gross — RCT/APT belong where raw noise symbols exist, no SP 800-90B claim on a conditioned 32-byte result	OPEN	02

Recovery/inheritance rehearsal with real cards, documents, devices, transactions	OPEN	05
Evidence level ≤ 2 (reference / cross-check) no host or target implementation has passed	SETTLED	all

Documents: 01 Blueprint · 02 QKEC-1 · 03 Standards · 04 PSBT Checklist · 05 Recovery Decision/Kit · 06 Card Protocol · 07 Recovery Document · 08 Component Map. "Settled" = selected product direction, not implemented evidence.

03 CONFORMING REFERENCE VALUES

The 2-of-3 spine – now the wallet, not a candidate

Built from three independent 24-word (100-roll) ceremonies; regenerated and cross-reproduced by embit. A **reference computation** for the selected policy — not a canonical descriptor to fund.

```
fingerprints A = 2fae9711 B = 72a14ab8 C = badaa8ff
wallet      wsh(sortedmulti(2, [A]/48'/0'/0'/2', [B]..., [C]...)) descriptor ...#kuyqfyj3
recv /0/0    bc1qecahkw0srs8xpmgwqya49qtknxwvkwr0ky4jds52scsw4etmd6s5s93pr
recovery representation: also store two ordinary receive/change descriptors (the conservative authority);
BIP-389 multipath is an optional Draft convenience, not the sole decades-long recovery record.
```

THE RECOVERY STOP-SHIP (ARCHITECTURE SELECTED, REHEARSAL OPEN)

The *architecture* now answers "what if one factor is lost?" — spend once with A1+C or B+C, then sweep into a completely fresh A/B'/C'/A2'/D' kit; a lost factor is never reissued. What is still **stop-ship** is proving the selected answer works with real cards, documents, devices, and transactions (the rehearsals in doc 05), plus the open A1 print-codec/capture, card-model, and QR-performance items. No material funding until those close. This register stays subordinate to QuietKey Core Architecture v1 and QK2-04; nothing here is incorporated into QK2-04 beyond the settled scope directions above.

